

EIE3123 DYNAMIC ELECTRONIC SYSTEMS: QUIZ 3

1. Consider the system shown in Figure Q1. Plot the root loci for the system. Determine the value of K such that the damping ratio of the dominant closed-loop poles is 0.5. Then determine all closed-loop poles. Plot the unit-step response curve with MATLAB.

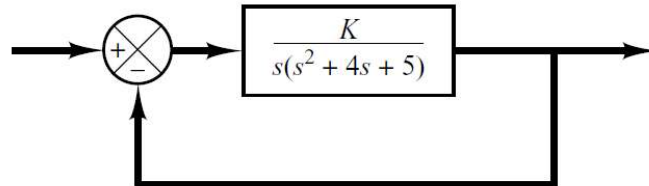


Figure Q1

2. Consider the control system shown in Figure Q2. Design a compensator such that the unit-step response curve will exhibit maximum overshoot of 30% or less and settling time of 3 sec or less.

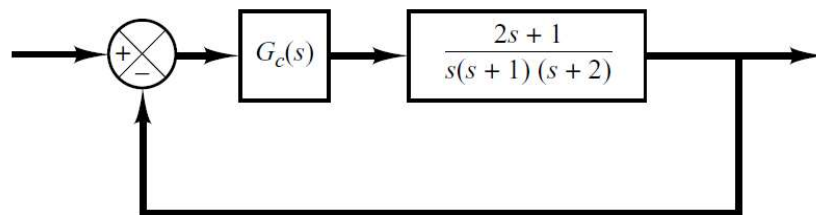


Figure Q2